

Unit 11, Lesson 7: Translating Between Representations of Proportional Relationships

Benchmarks

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

Learning Targets

- ☐ I can determine the constant of proportionality of a proportional relationship.
- ☐ I can translate a given representation of a proportional relationship to a written description, table, or equation.

11.7.1 Warm-Up: Equivalent Ratios

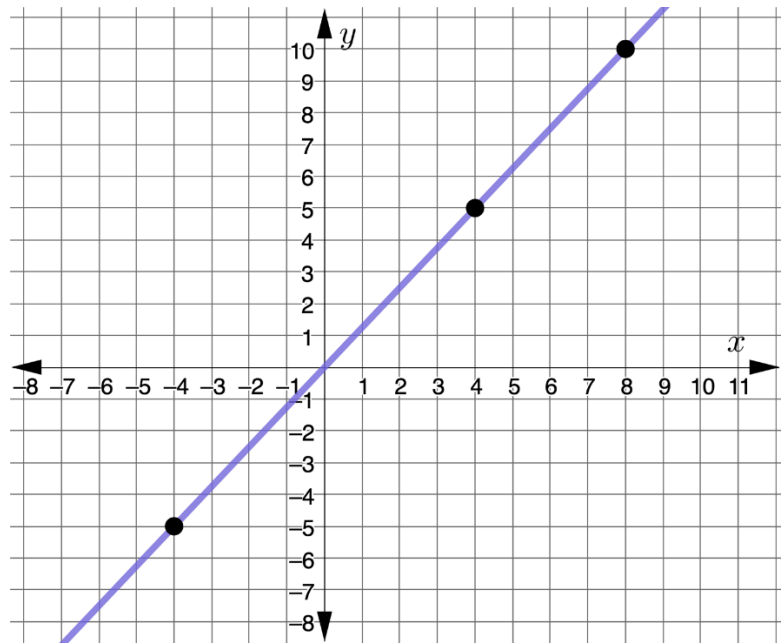
1. Fill in the blue boxes using the digits 1 to 9 to make two ratios that are equivalent to 20 : 30.

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11.7.2 Guided Instruction: Translating Between Different Representations of Proportional Relationships

The graph of a proportional relationship is shown.



1. Complete the representations of the proportional relationship by filling in the blanks in the chart.

Table of Values	Equation	Written Description										
<table><tr><th>x</th><th>y</th></tr><tr><td>-4</td><td></td></tr><tr><td>0</td><td></td></tr><tr><td>4</td><td></td></tr><tr><td>8</td><td></td></tr></table>	x	y	-4		0		4		8		$y = \boxed{} x$	As the value of x increases by _____, the value of y increases by _____.
x	y											
-4												
0												
4												
8												

2. Suppose in the previous relationship, x represents the price, in dollars, a store owner pays for merchandise from their manufacturer and y represents the price, in dollars, for which the store owner sells the merchandise.
- a. What is the constant of proportionality, $\frac{y}{x}$?

- b. Explain what the constant of proportionality means in this context.
- c. Complete the written description.
As the value of x increases by _____ dollars, the value of y increases by _____ dollars.
- d. Discuss with your partner what is different about the written description given the context compared to the written description in Problem 1.
- e. Explain which parts of the graph of this relationship do not make sense in the context.
- f. Use the equation to determine the sale price of an item the store owner purchases for \$32.60.
- g. Write the coordinate pair of the point on the graph of this relationship that represents the item described in Part F.
- h. Add a row to the table of values to represent this point.
- i. Show that the ratio $\frac{y}{x}$ for this point is equivalent to the constant of proportionality.

11.7.3 Guided Practice: Translating Between Different Representations of Proportional Relationships

1. Each table represents a proportional relationship.
- a. Find the constant of proportionality and write an equation that represents the relationship.

Table of Values	<i>s</i>	2	3	5	10
	<i>P</i>	8	12	20	40
Constant of Proportionality					
Equation	$P = \underline{\hspace{2cm}}$				

- b. Find the constant of proportionality and write an equation that represents the relationship.

Table of Values	<i>d</i>	2	3	5	10
	<i>c</i>	6.28	9.42	15.7	31.4
Constant of Proportionality					
Equation	$c = \underline{\hspace{2cm}}$				

2. It took Petra 5 minutes to fill a cooler with 8 gallons of water from a faucet flowing at a steady rate. Let w be the number of gallons of water in the cooler after t minutes.
- a. Circle the two equations that represent the relationship between w and t .
- $w = 1.6t$ $w = 0.625t$ $t = 1.6w$ $t = 0.625w$
- b. What does 1.6 represent in the context of this situation?
- c. What does 0.625 represent in the context of this situation?

3. In Citrus County, FL, a 6% sales tax is added to the price of any item sold.
- a. Complete the table of values to represent the relationship between an item’s price, p , in dollars, and the total cost to a customer, t , in dollars, after sales tax is applied.

p (\$)	1.50	4	20
t (\$)			

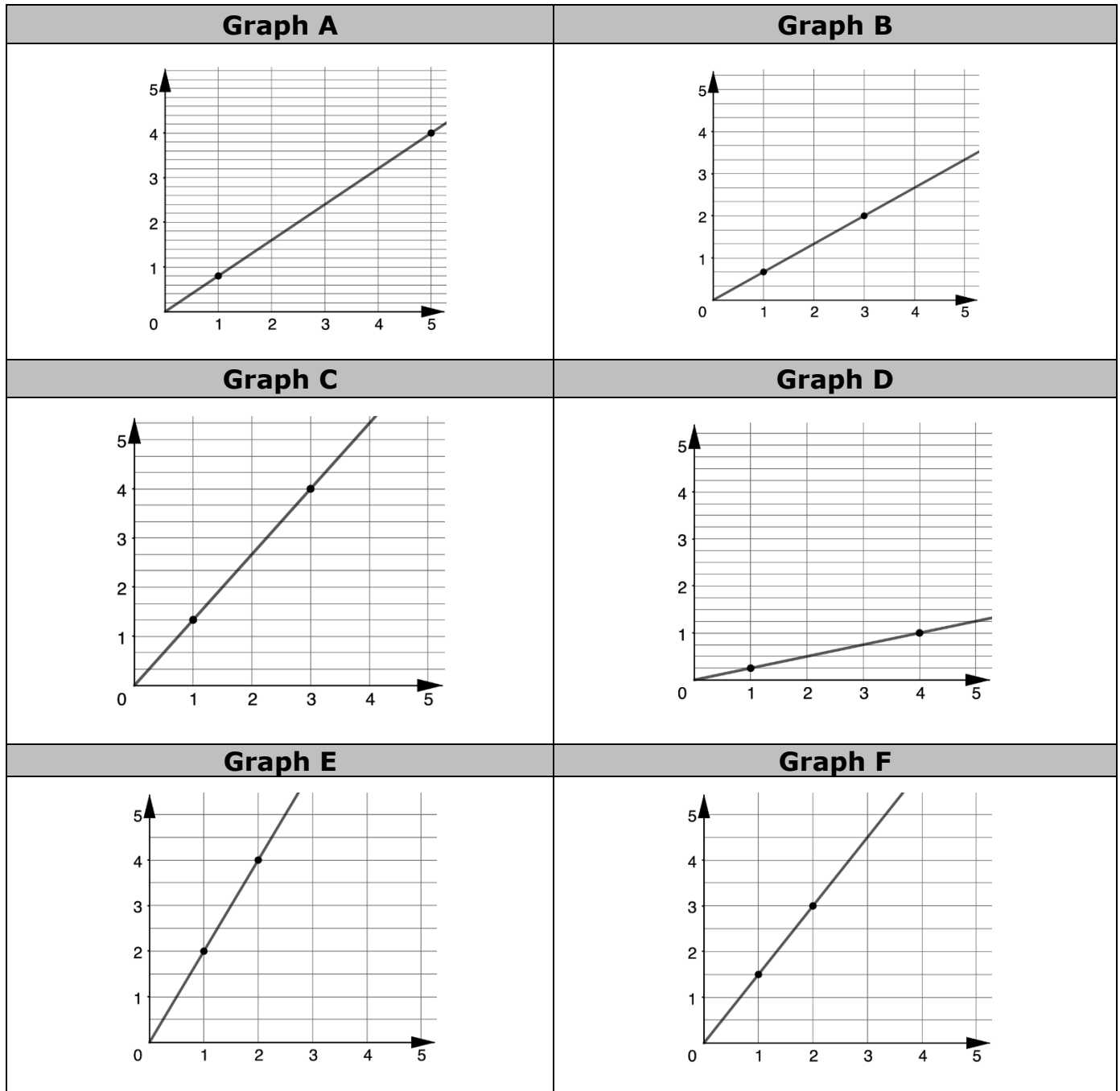
- b. Write an equation that could be used to find the total cost, t , in dollars, of any item sold in Citrus County, after taxes.

11.7.4 Collaboration: Matching Equations to Graphs

Several equations are shown.

$y = 2x$	$y = \frac{4}{5}x$	$y = \frac{1}{4}x$	$y = \frac{2}{3}x$	$y = \frac{4}{3}x$	$y = \frac{3}{2}x$
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- Work with your partner to match each equation to the graph representing the same relationship. Write the equation in the space provided below each graph.



2. Choose one graph and explain how you determined the equation that matched the graph.

3. Share your explanation with your partner. If necessary, make edits to your explanation based on feedback from your partner.

4. A store is having a 20% off sale. With your partner, discuss which graph represents the relationship between the original price, in dollars, of an item, x , and its sale price, y , in dollars. Then, write a brief summary of your discussion.

11.7.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on translating proportional relationships to written descriptions, tables, or equations?



